

CompositionMeasure for Approximate<MaxDivergence>

Michael Shoemate (@Shoeboxam)

This proof resides in “**contrib**” because it has not completed the vetting process.

Proves soundness of the implementation of **CompositionMeasure** for **Approximate<MaxDivergence>** in **mod.rs** at commit **f5bb719** (outdated¹).

1 Hoare Triple

Precondition

Compiler-verified

Types matching pseudocode.

Caller-verified

None

Pseudocode

```
1 class CompositionMeasure(ApproximateMaxDivergence):
2     def composability( #
3         self, adaptivity: Adaptivity
4     ) -> Composability:
5         return Composability.Concurrent
6
7     def compose(self, d_mids: Vec[tuple[Self_Distance, u32]]) -> Self_Distance:
8         eps_g, del_g = 0.0, 0.0
9         for (eps_i, del_i), k_i in d_mids:
10             k_i = f64.from_(k_i)
11             eps_g = eps_g.inf_add(eps_i.inf_mul(k_i))
12             del_g = del_g.inf_add(del_i.inf_mul(k_i))
13         return eps_g, del_g
```

Postcondition

Theorem 1.1. `composability` returns `Ok(out)` if the composition of a vector of privacy parameters `d_mids`, where each `d_mid` is charged `k` times, is bounded above by `self.compose(d_mids)` under `adaptivity` `adaptivity` and `out-composability`. Otherwise returns an error.

Proof. By the postconditions of **InfAdd** and **InfMul** we have that $\sum_i k_i d_{mids_i} \leq \text{compose}(d_mids)$, where the summation is applied independently to epsilons and deltas, and the comparison applies to both the global epsilon and global delta.

¹See new changes with `git diff f5bb719..d7b4364b rust/src/combinators/sequential_composition/mod.rs`

Adaptivity	Sequential	Concurrent
Non-Adaptive	Theorem 1[DKM ⁺ 06]	Theorem 1.3[VZ23]
Adaptive	Theorem 1[DKM ⁺ 06]	Theorem 1.3[VZ23]
Fully-Adaptive	Lemma 3.5[RRUV21]	Theorem 1.3[VZ23]

The reference for fully adaptive sequential composition is specifically for the version of the paper prior to correction (v1, not v2). The correction is not necessary for the privacy proof when odometers are defined in terms of a truncated view.

This table is reflected in the implementation of `composability` on line 2.

□

References

- [DKM⁺06] Cynthia Dwork, Krishnaram Kenthapadi, Frank McSherry, Ilya Mironov, and Moni Naor. Our data, ourselves: Privacy via distributed noise generation. In Serge Vaudenay, editor, *Advances in Cryptology - EUROCRYPT 2006*, pages 486–503, Berlin, Heidelberg, 2006. Springer Berlin Heidelberg.
- [RRUV21] Ryan Rogers, Aaron Roth, Jonathan Ullman, and Salil Vadhan. Privacy odometers and filters: Pay-as-you-go composition, 2021.
- [VZ23] Salil Vadhan and Wanrong Zhang. Concurrent composition theorems for differential privacy. In *Proceedings of the 55th Annual ACM Symposium on Theory of Computing*, STOC '23, page 507–519. ACM, June 2023.